

CHAPTER 4: LIGHT

Important Terms & Definitions

1. Nature & Propagation of Light

Light: A form of energy that produces the sensation of vision. It travels at a speed of $3 \times 10^8 \text{ m s}^{-1}$ in air or vacuum.

Rectilinear Propagation of Light: The property of light travelling in a straight-line path in a homogeneous medium.

Ray of Light: A line segment with an arrow indicating the direction of propagation of light.

Beam of Light: A group of adjacent light rays. It may be parallel, converging (rays meeting at a point), or diverging (rays spreading from a point).

2. Sources of Light & Optical Media

Luminous Bodies: Bodies that emit their own light, e.g., the Sun and electric bulbs.

Non-luminous Bodies: Bodies that do not emit light on their own and are seen by reflecting light, e.g., the Moon.

Bioluminescence: Light emitted naturally by living organisms.

Hot Sources: Sources that emit light when extremely hot, e.g., the Sun and filament bulb.

Cold Sources: Sources that emit light while remaining relatively cool, e.g., glow-worms.

Point Source: A light source whose size is very small, approximately the size of a pinhead.

Extended Source: A light source that is larger than a point source.

Optical Medium: Any material through which light can partially or wholly pass.

Homogeneous Medium: A medium having uniform composition throughout.

Heterogeneous Medium: A medium whose composition varies at different points.

Transparent Medium: A medium that allows light rays to pass through completely, e.g., clear glass and water.

Opaque Medium: A medium that does not allow light rays to pass through, e.g., wood and metal.

Translucent Medium: A medium that allows light to pass through only partially, e.g., oil paper and fog.

3. Shadows & Eclipses

Shadow: The dark region formed behind an opaque body placed in the path of light due to rectilinear propagation.

Umbra: The central, completely dark region of a shadow.

Penumbra: The partially dark region surrounding the umbra.

Eclipses: Phenomena that occur when celestial bodies cast their shadows on one another.

Lunar Eclipse: Occurs when the Earth's shadow falls on the Moon.

Solar Eclipse: Occurs when the Moon's shadow falls on the Earth.

Pinhole Camera: A simple device that forms an inverted image based on the rectilinear propagation of light.

Magnification (m): The ratio of image size to object size, or image distance to object distance, as applicable.

4. Reflection of Light

Reflection: The bouncing back of light rays when they strike a polished surface.

Regular Reflection: Reflection from a smooth surface in a well-defined direction.

Diffused Reflection: Reflection in which light is scattered in different directions from a rough surface.

Law of Reflection 1: The incident ray, reflected ray and the normal at the point of incidence all lie in the same plane.

Law of Reflection 2: The angle of incidence ($\angle i$) is equal to the angle of reflection ($\angle r$).

5. Images & Optical Devices

Real Image: An image formed when reflected rays actually intersect. It can be caught on a screen and is generally inverted.

Virtual Image: An image formed when reflected rays appear to diverge from a point. It cannot be caught on a screen and is generally erect.

Lateral Inversion: The phenomenon in which the right side of an object appears as the left side in its plane-mirror image.

Kaleidoscope: An instrument that uses multiple reflections between inclined plane mirrors to create changing patterns.

Periscope: An optical device consisting of mirrors or prisms, usually inclined at 45° , used to observe objects not in the direct line of sight.

6. Spherical Mirror Terminology

Concave Mirror (Converging Mirror): A spherical mirror having an inner reflecting surface.

Convex Mirror (Diverging Mirror): A spherical mirror having an outer reflecting surface.

Pole (P): The geometric centre of the reflecting surface of a spherical mirror.

Centre of Curvature (C): The centre of the sphere of which the spherical mirror forms a part.

Radius of Curvature (R): The radius of the sphere of which the mirror forms a part.

Principal Axis: The straight line passing through the pole (P) and centre of curvature (C).

Principal Focus (F): The point on the principal axis where rays parallel to the principal axis meet after reflection in a concave mirror, or appear to diverge from in a convex mirror.

Focal Length (f): The distance between the pole and principal focus. For a spherical mirror, $f = R/2$.

Aperture: The diameter of the reflecting surface of a spherical mirror.

7. Dispersion & Light Intensity

Dispersion of Light: The splitting of white light into its seven constituent colours (VIBGYOR) when it passes through a prism.

Candela: The SI unit of luminous intensity.

Candle Power: A measure comparing the intensity of a light source with that of a standard candle.

Lumen: A derived unit of luminous flux. According to the supplied reference, 1 lumen = 12.56 candle power.

Lux / Foot-candle: Units used to measure illuminance. According to the supplied reference, 1 foot-candle = 10.76 lux.

Key Relations to Remember

- Speed of light in air/vacuum = $3 \times 10^8 \text{ m s}^{-1}$
- For a spherical mirror: $f = R/2$
- Law of reflection: $\angle i = \angle r$
- 1 foot-candle = 10.76 lux
- VIBGYOR = Violet, Indigo, Blue, Green, Yellow, Orange, Red